Transients on Transmission Lines – Part 2

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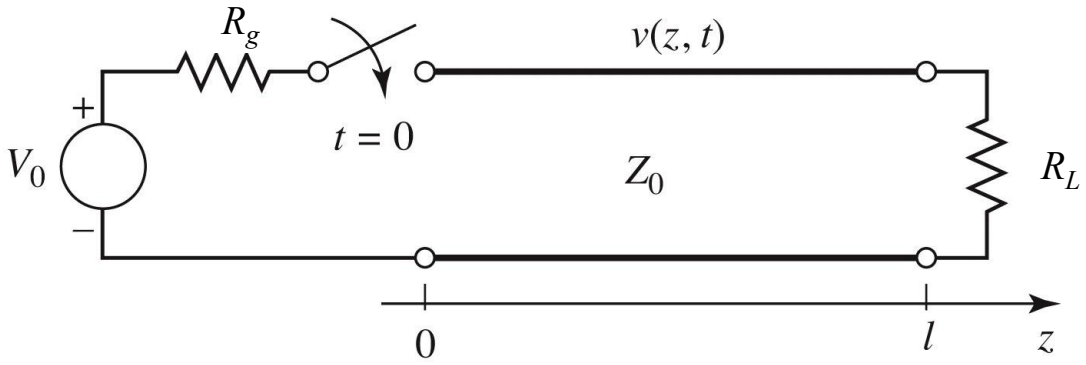
Lecture Objective

- By the end of this lecture, you will be able to:
  - Analyze the time-domain response for a transmission line terminated in arbitrary resistive generator and load impedances.

Transient Response of a Transmission Line Terminated in Arbitrary Generator and Load Resistances



(a)

Figure 2.21a

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2. Transient Response of a Transmission Line Terminated in Arbitrary Generator and Load Resistances

- At time $t = 0$, a disturbance (the leading edge of the pulse) is excited on the transmission line.
- The amplitude of the disturbance is given by voltage division between the generator impedance and the characteristic impedance of the line:

$$V^+ = \frac{Z_0}{R_g + Z_0} V_0$$

- This disturbance propagates down the transmission line in the +z-direction with velocity v_p until it encounters the load where some/all/none of it is reflected.

3. Transient Response of a Transmission Line Terminated in Arbitrary Generator and Load Resistances

- At time $t = T$, the disturbance (the leading edge of the pulse) is reflected from the load and starts traveling back toward the generator.
- The amplitude of the disturbance is given by

$$V^- = \Gamma_L V^+ = \frac{R_L - Z_0}{R_L + Z_0} \frac{Z_0}{R_g + Z_0} V_0$$

- This disturbance propagates down the transmission line in the -z-direction with velocity v_p until it encounters the generator where some/all/none of it is reflected.

4. Transient Response of a Transmission Line Terminated in Arbitrary Generator and Load Resistances

- At time $t = 2\ T$, the disturbance (the leading edge of the pulse) is reflected from the generator and starts traveling back toward the load.
- The amplitude of the re-reflected disturbance is given by

$$V^{-+} = \Gamma_g V^- = \Gamma_g \Gamma_L V^+$$

- This disturbance propagates down the transmission line in the +z-direction with velocity v_p until it encounters the load where some/all/none of it is reflected.

5. Transient Response of a Transmission Line Terminated in Arbitrary Generator and Load Resistances

- For $t > 2T$ (after the disturbance returns to the generator), we must have that

$$v(0, t > 2T) = V_0 - R_g\, i(0, t > 2T)$$

- Where

$$v(0, t > 2T) = V^+ + V^- + V^{-+}$$

Re-reflected wave

$$i(0, t > 2T) = \frac{1}{Z_0}\left(V^+ - V^- + V^{-+}\right)$$

6. Transient Response of a Transmission Line Terminated in Arbitrary Generator and Load Resistances

- Solving for the voltage reflection coefficient at the generator:

$$\cancel{V^+} + V^- + V^{-+} = V_0 - \frac{R_g}{Z_0}(\cancel{V^+} - V^- + V^{-+})$$

$$\frac{Z_0}{R_g + Z_0}V_0 \qquad\qquad \frac{Z_0}{R_g + Z_0}V_0$$

$$\Gamma_g = \frac{V^{-+}}{V^-} = \frac{R_g - Z_0}{R_g + Z_0}$$

7. Transient Response of a Transmission Line Terminated in Arbitrary Generator and Load Resistances

- This process of bouncing back and forth of waves goes on indefinitely until steady state is reached.
- In DC steady state, the (lossless) transmission line behaves like an ideal interconnect, and so we have

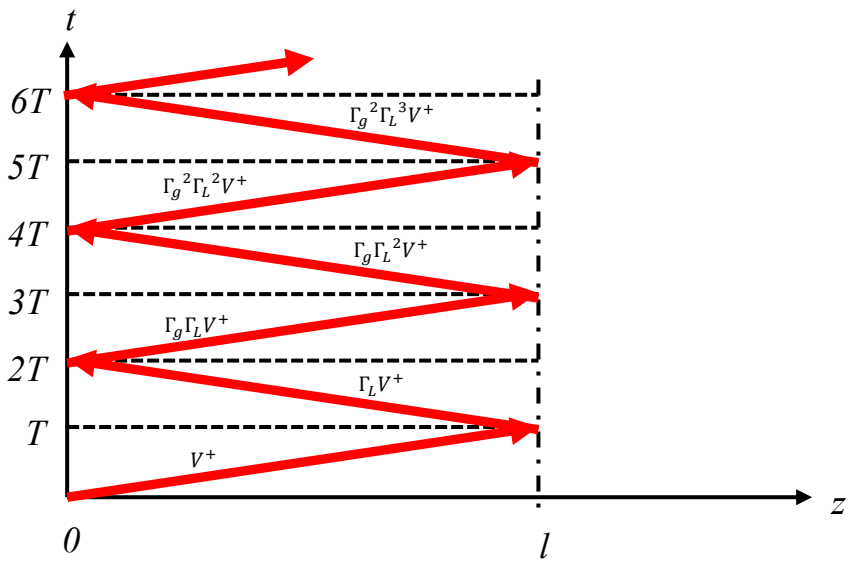
$$v\left(z,t\to\infty\right)=\frac{R_L}{R_g+R_L}V_0$$

$$i\left(z,t\to\infty\right)=\frac{V_0}{R_g+R_L}$$

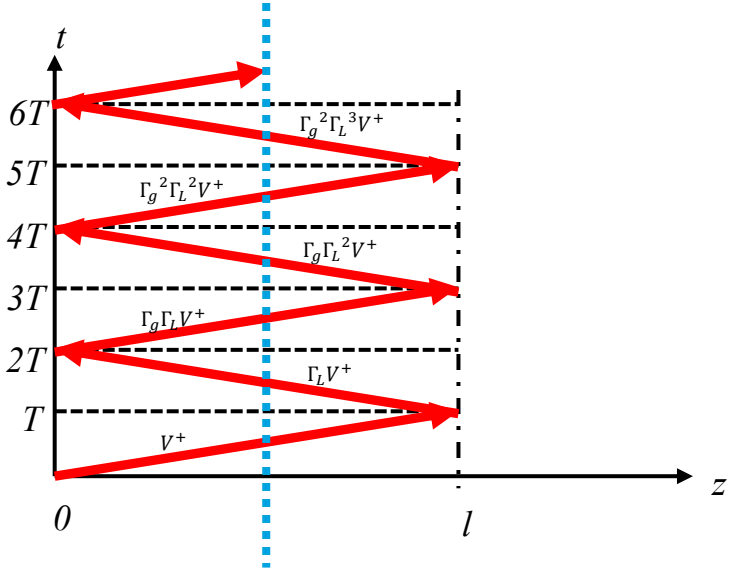
Bounce Diagrams

- To keep track of transient behavior on a transmission line, we can make use of a ***bounce diagram*** (*a.k.a.* a ***reflection diagram*** or a ***space-time diagram***).
- In most situations, we use a <u>voltage</u> bounce diagram, but it is also possible to have a <u>current</u> bounce diagram.

2. Bounce Diagrams



Compute Voltage at Fixed Location Versus Time Using Bounce Diagram



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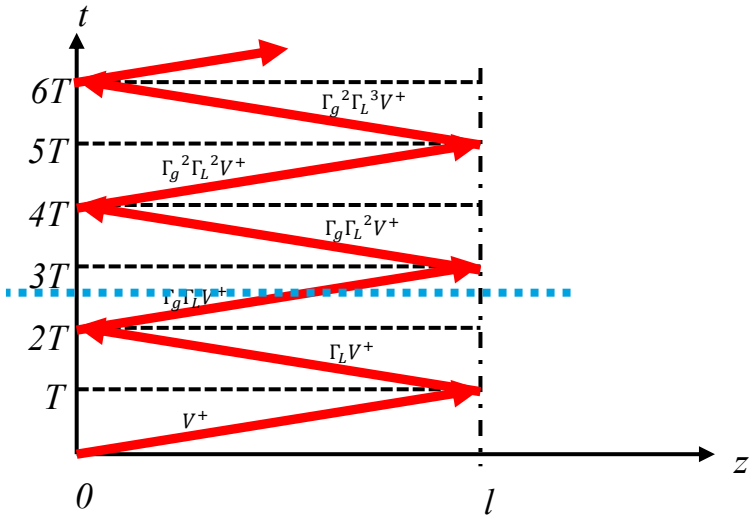


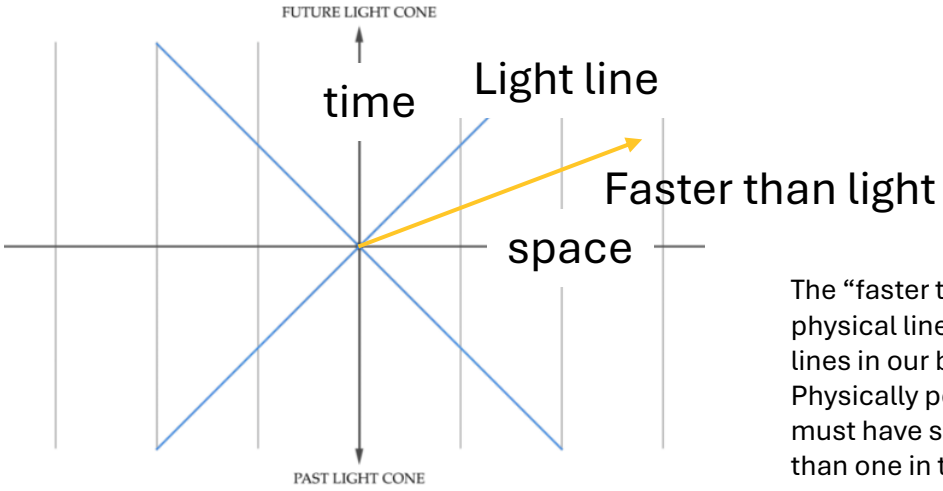
DC Steady-State Voltage

$$v(z,t\to\infty) = V_0\frac{Z_0}{R_g+Z_0}\left(1+\Gamma_L+\Gamma_L\Gamma_g+{\Gamma_L}^2\Gamma_g+{\Gamma_L}^2{\Gamma_g}^2+\cdots\right)$$
$$= V_0\frac{Z_0}{R_g+Z_0}(1+\Gamma_L)\left\{1+\Gamma_L\Gamma_g+\left(\Gamma_L\Gamma_g\right)^2+\cdots\right\}$$
$$= V_0\frac{Z_0}{R_g+Z_0}(1+\Gamma_L)\sum_{n=0}^{\infty}\left(\Gamma_L\Gamma_g\right)^n$$

$$v(z,t\to\infty) = V_0\frac{Z_0}{R_g+Z_0}(1+\Gamma_L)\sum_{n=0}^{\infty}\left(\Gamma_L\Gamma_g\right)^n$$
$$= V_0\frac{Z_0}{R_g+Z_0}(1+\Gamma_L)\frac{1}{1-\Gamma_L\Gamma_g} \qquad = V_0\frac{R_L}{R_L+R_g}$$

Question

- In physics, particularly in discussions of special relativity, we often use space-time diagrams. We also hear that the speed of light is represented by lines with slope 1 and that things that could have affected an event must lie in the “past light cone” and things that can be affected by an event must lie in the “future light cone”. An example of such a space-time diagram is shown below. The question is: in our space-time or bounce diagrams, the lines we draw have slopes less than 1, and yet we know we are not traveling faster than light speed; what is going on there? (In other words, what do the physicists do differently in their space-time diagrams?)



The “faster than light” non-physical line looks like the lines in our bounce diagram. Physically possible lines must have slopes greater than one in this diagram.

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